



Anaerobic Co-Digestion of Tannery Wastes and Untreated/Pretreated Oat Straw

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Abstract

The combination of carbon and nitrogen-rich co-substrates results in a better balance and increases the stability of the anaerobic co-digestion (AcoD) process. In this work, the AcoD process of waste from the leather industry (shavings and sludge) with waste from agriculture (oat straw) was assessed with regard to the energy (biogas production) and waste treatment efficiency (reduction of organic matter). The results indicate that the addition of untreated oat straw improved the AcoD process, increasing biogas production (25.44 mL of cumulative biogas/g of VSS added) by almost 60% when compared to the AcoD of only leather waste (16.17 mL/gVSS). Also, the effect of acid, alkaline, thermal, and the combination of these pretreatment techniques was evaluated on the lignocellulosic composition of oat straw and on methane yields. Pretreatments improved the characteristics and bioavailability of oat straw, particularly in methodologies that use alkali, with a significant increase in cellulose content accompanied by a decrease in hemicellulose and lignin content. However, the possible formation of secondary products or sterilization of important microorganisms did not reflect in a greater production of biogas: 21.06 mL/gVSS for oat straw pretreated only with HCl and 21.91 mL/gVSS for oat straw pretreated with HCl in autoclave; 5.20 mL/gVSS for oat straw pretreated with NaOH; and 3.43 mL/gVSS for oat straw pretreated with NaOH in autoclave; thermal pretreatment probably has generated toxic compounds from hemicellulose and cellulose degradation, which inhibited the AcoD process and, as consequence, virtually no biogas was produced.

Keywords Renewable energy · Anaerobic digestion · Lignocellulosic biomass · Leather waste · Pretreatment · Methane

Introduction

Renewable energy production methods have been examined to provide accessible and sustainable energy sources which are essential for economic growth, increasing energy security, and reducing the risk of climate change [1]. Worldwide renewable energy consumption is expected to increase by 3% per year between 2018 and 2050 and becomes the leading source of primary energy consumption by 2050 [2]. Among renewable forms of energy, the use of waste energy is one of

the most attractive for treating waste while producing energy from it.

Anaerobic digestion (AD) is a viable alternative to renewable energy, which can be described as a process in which biomass resources are submitted to a microbial process under anaerobic conditions to produce two major products: energy-rich biogas and nutrient-rich digested residues [3]. Biogas production is independent of seasonal fluctuations, can be stably produced, and, therefore, promises a reliable way to produce energy [1]. Despite producing alternative fuels and high-quality fertilizer, the biogas production technology offers multiple benefits like complete waste recycling, greenhouse gas reduction, and environmental protection from pollutants [4].

The production of biogas from lignocellulosic biomass creates a good opportunity to convert vast biomass resources into renewable energy [3]. Unlike purpose-grown energy crops, the residue of lignocellulosic material (straw) does not compete with food production. The grain is seen as the main product and much of the straw generated worldwide

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is left in the field after harvesting. Straw is therefore often available in large quantities at a low price [5].

Lignocellulosic material is mainly composed of three different types of polymers: cellulose, hemicellulose, and lignin. While cellulose has a rigid and crystalline form, hemicellulose has a lower molecular weight and short lateral chains, which corresponds to an easy hydrolyzable polymer. The third polymer, lignin, is composed mainly of phenolic compounds which binds together hemicellulose and cellulose and runs throughout the three-dimensional structure providing a protective sheath over hemicellulose and cellulose, giving resistance to microbial attack. Hemicellulose and cellulose, collectively called holocellulose, are polymers of sugars which are biodegradable and can be effectively utilized to produce biogas [6, 7]. The presence of lignin is a hindrance to the enzymatic hydrolysis of cellulose during the anaerobic digestion, what makes the hydrolysis of lignocellulose often becomes the rate-limiting step during process [8, 9]. In fact, the major difficulty with the lignocellulosic substrate is the proper digestion of biomass during the AD due to the complex and recalcitrant nature of feedstock, which makes absolutely necessary a pretreatment step [10, 11].

Different pretreatment methods have been applied to eliminate physical and chemical barriers, making cellulose accessible for enzymatic hydrolysis [4, 12]. The selection of the optimum pretreatment technique depends on multiple factors such as the characteristics of the lignocellulosic material (crystallinity of the lignocellulose, the degree of polymerization, accessible surface area), the capital and operating cost of the pretreatment, and the ease of the operation [1, 13]. There are several studies reported in the literature with methodologies for chemical and thermal pretreatments of lignocellulosic materials, resulting in changes in the lignocellulosic composition of these materials and influence on the performance of subsequent anaerobic biological processes.

Gomez-Tovar et al. [11] reported the acid pretreatment (2% HCl, 90 °C) of oat straw as part of a sequential pretreatment (acid/alkaline/enzymatic) to convert the lignocellulosic material in soluble sugars and observed that 85.5% of the hemicellulose and 30.3% of the cellulose were hydrolyzed in the acid step. Mancini et al. [14] investigated the effect of alkaline pretreatment (NaOH at 30 °C for 24 h) on the biogas production from the anaerobic digestion of wheat straw and observed that the cumulative biomethane production yield of the untreated feedstock was enhanced by 15% by alkaline pretreatment. Similarly, rice straw was pretreated with NaOH (0.5%, 1%, 1.5%, and 2% w/v for 3 h at 37 °C) and, as compared to raw rice straw, maximum biogas and methane enhancement of 50 and 71%, respectively, was observed for 1.5% w/v NaOH pretreated rice straw [7]. Rajput et al. [15] studied the effect of thermal pretreatment (120, 140, 160,

and 180 °C for 60 min) on anaerobic digestion of wheat straw and found a 53% increase in biogas yield for wheat straw pretreated at 180 °C as compared to untreated. Ferreira et al. [6] evaluated the biochemical methane potential of steam-exploded wheat straw under different temperature–time combinations (150, 220 °C and 1–15 min) and reported that the optimum was obtained for 1 min and 220 °C, resulting in a 20% increase in methane production with respect to non-treated straw. They also observed that for more severe pretreatment conditions, the biodegradability of wheat straw decreased due to a possible formation of inhibitory compounds. Chandra et al. [16] presented the results of an experimental methane fermentation study on untreated, NaOH (4% at 37 °C for 120 h) and hydrothermal (200 °C for 10 min) pretreated substrates of wheat straw which produced, respectively, 111.6% and 20% higher methane compared to the untreated wheat straw substrate.

The combination of chemical and thermal pretreatments has also been investigated. Taherdanak et al. [17] evaluated the improvement of biomethane production from wheat submitted to pretreatment with H₂SO₄ 1% (v/v) at room temperature for 10 min followed by autoclave at 121 °C for four different time (10, 30, 60, and 120 min) and achieved an increase in methane yield of 8.9% after pretreatment for 60 min and 15.5% after 120 min, compared to untreated wheat. Patowary and Baruah [8] investigated the combination of chemical (banana peel ash and calcium hydroxide) and thermal treatments (60–90 °C for time intervals of 2, 6, and 10 h) of rice straw and corn stalk and reported that biogas production of rice straw and corn stalk pretreated at 90 °C for 6 h was enhanced by 62% and 66%, respectively as compared to untreated rice straw and corn stalk. Kim et al. [18] assessed the effects of autoclaving (121 °C, 1.45 atm, 60 min) of rice straw after addition of 2% H₂SO₄ and the results showed a reduction of about 20% in methane yield, indicating the inhibitory effect of the acid in the biogas production process.

In addition to structural features, lignocellulosic wastes have a high carbon/nitrogen (C/N) ratio which limits its efficient use in AD processes for methane production [19]. For example, oat straw, the biomass used in this study, has a C/N ratio around 50, higher than the optimal range for AD (20–30) [20]. To overcome this limitation, lignocellulosic biomass is often co-digested with nitrogen-rich feedstock, in order to maintain the optimum C/N ratio and increase the stability of the process [21]. The leather industry generates substantial quantities of nitrogen-rich solid waste [22] and could be a good option to balance the C/N ratio of AcoD with oat straw. Leather shavings and sludge from wastewater treatment plants (WWTP) are the main solid wastes generated by tanneries [23] and AD has become an attractive solution in the perspective of sustainable and integrated management of these wastes [24, 25]. Additionally,

the combination of co-substrates results in other positive effects to the system, such as a better balance (pH, moisture), dilution of potentially toxic compounds, and supplementation of trace elements. Thus, AcoD of lignocellulosic and tannery residues is a possible way to solve the limitations of mono-digestion, increase biogas production, improve waste recycling, and contribute to environmental protection [26], furthermore to the economic advantage of sharing the AD system and treating more than one waste at the same time [20, 21].

The novelty of this work is the possibility of co-digesting waste from agricultural cultivation and from leather industry, in order to maintain a balanced C/N ratio during the AcoD process. There are no reported studies involving AcoD of leather shavings, sludge from WWTP of tanneries, and oat straw. This work also aimed to compare the effects of acid, alkaline, thermal, and a combination of these pretreatment techniques on the lignocellulosic composition of oat straw and on the methane yields from its subsequent AcoD. The AcoD process was assessed with regard to the energy (biogas production) and waste treatment efficiency (reduction of organic matter).

Materials and Methods

Substrates and Inoculum

The leather shavings from tannery factories that use chromium as tanning agent and vegetable tannin as retanning agent and the sludge (inoculum) of WWTP were collected in tanneries, located close to Porto Alegre, Rio Grande do Sul (Brazil). Oat (*Avena sativa*) straw was harvested from agricultural fields in the north of Rio Grande do Sul (Brazil).

Pretreatments of Oat Straw

The whole oat straw sample that was used in this work was previously oven dried at 60 °C, ground in a knife mill, and passed through a 6-mesh sieve. This initial preparation was not a variable evaluated in this study, so that “untreated oat straw” corresponds to dry and ground oat straw, but not subjected to any of the pretreatments evaluated in this study. The strategy proposed for the previous hydrolysis of the oat straw consisted of acid, alkaline, and thermal pretreatments and the combination of those. Figure 1 shows the experimental flow adopted.

The chosen pretreatment conditions were based on previous studies [11, 14, 17] performed on lignocellulosic materials and which helped to increase the bioavailability of these materials. Acid hydrolysis was carried out in 10 g of oat straw that was suspended in 0.1 L of HCl (2%) at 90 °C for 2 h [11]. Alkaline pretreatment was carried out

in 10 g of oat straw that was suspended in 0.1 L of NaOH (1.6%) at 30 °C for 24 h [14]. After each pretreatment, the suspension was filtered and the liquid fraction was discarded, whereas the remaining fiber was washed with distilled water until neutral pH (7.0) was reached. Finally, the fiber was dried at 60 °C and stored. Thermal pretreatment was carried out by autoclaving (vertical autoclave, Phoenix Luferco) the untreated oat straw at 121 °C (1 atm) for 15 min [17].

The combinations of thermal and chemical pretreatments were also studied [17]. Acid + thermal pretreatment was carried in 10 g of untreated oat straw immersed in 0.1 L of HCl (2%) and autoclaved at 121 °C (1 atm) for 15 min. Alkaline + thermal pretreatment was carried in 10 g of untreated oat straw immersed in 0.1 L of NaOH (1.6%) and autoclaved at 121 °C (1 atm) for 15 min. After each pretreatment, the straw was filtered, washed to neutral pH, and oven dried (60 °C).

Biodigestion Tests

Biodigestion tests were performed in 300-mL hermetically sealed bench scale bioreactors, built in cylindrical glasses, containing an upper valve, to measure the volume of biogas produced, a lateral screw valve with septum, for the collection of samples for chromatographic analysis of the composition of biogas and a lower thread septum valve for liquid sample collection. The composition of the seven conditions studied is presented in Fig. 1. The ratio of sludge/shavings of 25 mL/1 g is equivalent, in proportion, to the amount of such waste generated by the tanneries of the region south of Brazil, according to local research [22]. A total of 200 mL of nutrient solution (2 g/L of yeast extract, 7 g/L of K₂HPO₄, and 3 g/L of KH₂PO₄) was employed to ensure favorable conditions for the growth and metabolism of the microorganisms [24]. The bioreactors were maintained at 35 °C (mesophilic temperature range) in a DL-CBE18L (DeLeo) bacterial culture oven. After 274 days of incubation, the bioreactors were opened and the precipitated and dissolved samples were collected. To analyze the composition of the biomass before biodegradation, the same tests were carried out in closed bottles, shaken for 1 h at room temperature on an ACB Labor Wagner shaker, and stored for 24 h at 4 °C for sedimentation before opening and the collection of liquid and solid samples [22]. All tests were performed in duplicate.

Statistical analysis was performed using the Microsoft Excel (version 2010) software for Windows. Statistical significance was evaluated using one-way ANOVA analysis of variance, with a Tukey post hoc test. Differences were accepted as significant when *p*-value < 0.05.

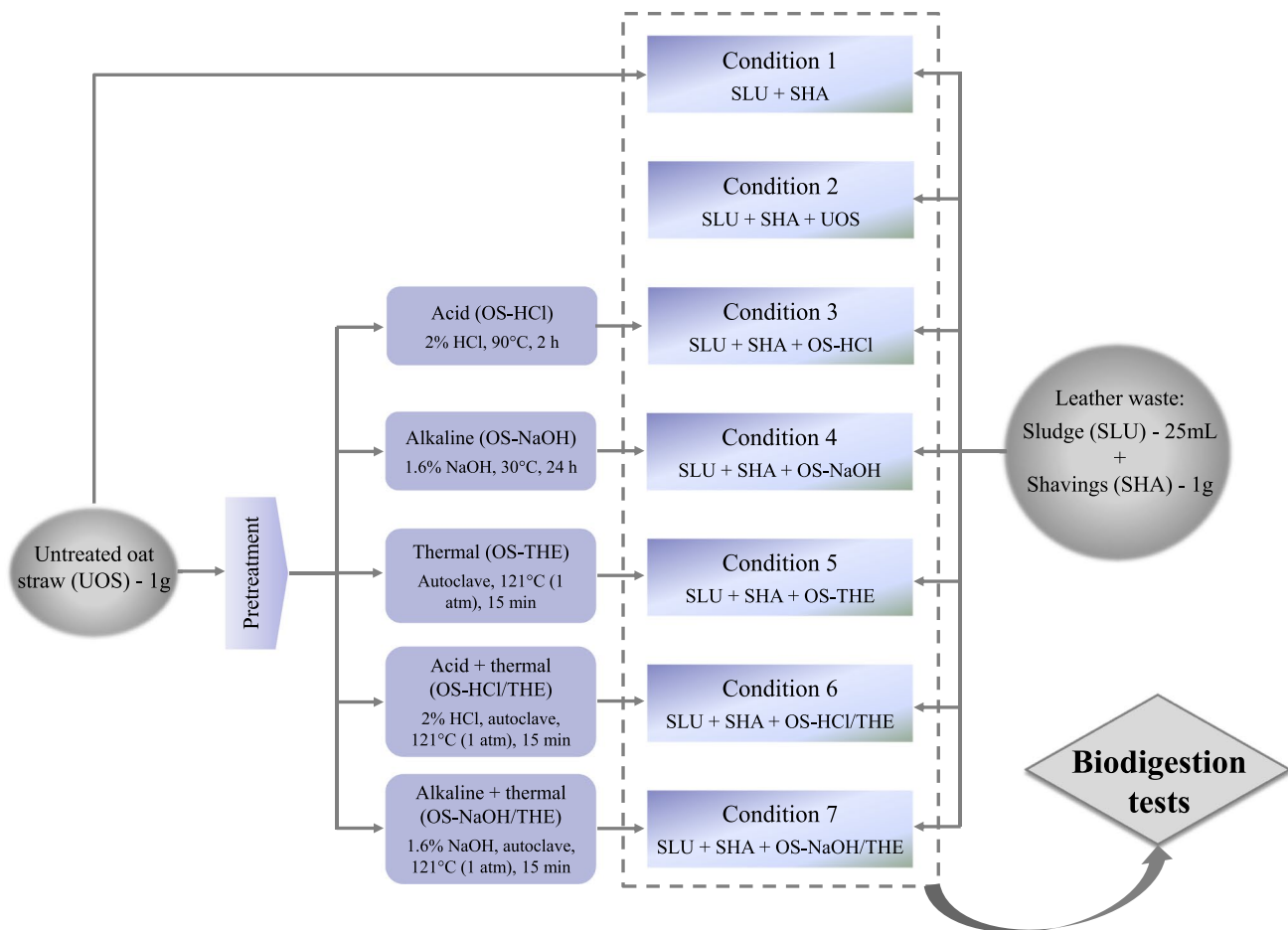


Fig. 1 Schematic of the experimental flow used in this study and composition of each reactor. SLU: sludge; SHA: shavings; UOS: untreated oat straw; OS-HCl: oat straw submitted to acid pretreatment; OS-NaOH: oat straw submitted to alkaline pretreatment; OS-

THE: oat straw submitted to thermal pretreatment; OS-HCl/THE: oat straw submitted to acid + thermal pretreatments; OS-NaOH/THE: oat straw submitted to alkaline + thermal pretreatments

Analytical Parameters

Waste Characterization

The wastes were characterized according to the following techniques. Gravimetric method (NBR ISO 4684:2014) was used to determine volatile solids (VS), using an analytical balance model EEQ9003F-B (Edutech), a drying oven (DeLeo), a muffle furnace (model Q318M/Quimis), and porcelain crucibles B-47 (Chiarotti). The determination of total organic carbon (TOC) and inorganic carbon (IC) was performed in a total organic carbon analyzer: TOC-VCSH Shimadzu for the liquid samples and SSM-500A Shimadzu, for the solids. Total Kjeldahl nitrogen (TKN) analysis was performed according to ASTM D3590-11, for liquid samples, and according to adapted D2868-10, for solid samples. The digestion of the samples was performed in a Velp Scientifica digester equipment, model DK20. The digested sample was distilled in

a nitrogen distiller equipment (Velp Scientifica) and the determination of the distilled ammonia was performed by titration.

Vegetable tannin concentration was measured in a T80 + UV/Vis Spectrometer (PG Instruments) at 278 nm with a calibration curve prepared with standard solutions of tannin of acacia (0.01–0.1 g/L). Total phenols were estimated as tannic acid equivalents, according to the Folin-Ciocalteu assay [27]. The absorbance of the samples was measured at 760 nm (T80 + UV/Vis Spectrometer by PG Instruments) and compared to the tannic acid calibration curve (0.01–0.1 g/L) elaborated in the same manner. For both analyses, vegetable tannin and total phenols, the sludge samples were diluted 100 and 10 times, respectively. Chromium (III) oxide concentrations were determined with ABNT NBR 13,341 (2010) method for residual bath. For the shavings, 1 g of the waste was dissolved in HCL 50% until total disintegration. pH was determined with a Digimed pH meter (DM-22). For pH measurement of

solid wastes, 2.5 g of shavings/oat straw was stirred for 24 h in 50 mL of distilled water.

Analysis of the Pretreatment Efficiency in the Composition of Oat Straw

Fourier transform infrared (FTIR) spectrophotometer (Frontier, Perkin Elmer) was used to analyze changes in the functional groups of oat straw before and after pretreatments. The spectra were formed from 4000 to 500 cm^{-1} with 32 times scanning.

Lignin, hemicellulose, and cellulose contents of oat straw before and after each pretreatment were measured based on chemical method [15, 28]. First, the amount of extractives in the biomass was determined by using solvent extraction, for which, 60 mL of acetone was used for 1 g of dried oat straw sample (G_0 , g) at a constant temperature of 90 °C for 2 h. After that, biomass sample was oven dried at 105 °C until constant weight was achieved (G_1 , g). For determining hemicellulose content, 150 mL of NaOH solution 20 g/L was added to 1 g of extractive-free dried biomass (G_1 , g). The mixture was boiled for 3.5 h with recycled distilled water, filtered, and washed until the value of pH approached 7. The sample was then dried to constant weight, cooled to room temperature in a desiccator, and weighted (G_2 , g). Lignin was determined by using 30 mL of sulfuric acid 72% for each gram of extractive-free dried biomass sample (G_3 , g). The mixture was kept at 8–15 °C for 24 h and transferred into a flask, diluted with 300 mL of distilled water, and boiled for 1 h with recycled distilled water. The mixture was then filtered, and the residues were washed until there was no more sulfate ion in the filtrate (detected by 10% barium chloride solution). The residue was then dried to a constant weight, cooled to room temperature in a desiccator, and weighted (G_4 , g). The extractive W_1 (wt%, db), hemicellulose W_2 (wt%, db), and lignin W_3 (wt%, db) were calculated according to Eqs. 1, 2, and 3, respectively. Cellulose content W_4 (wt%, db) was calculated by difference method as it was assumed that extractives, lignin, hemicellulose, and cellulose are the only components in biomass (Eq. 4).

$$W_1(\text{wt.}\%, \text{db}) = \frac{G_0 - G_1}{G_0} \times 100\% \quad (1)$$

$$W_2(\text{wt.}\%, \text{db}) = \frac{G_1 - G_2}{G_0} \times 100\% \quad (2)$$

$$W_3(\text{wt.}\%, \text{db}) = \frac{G_4(1 - W_1)}{G_3} \times 100\% \quad (3)$$

$$W_4(\text{wt.}\%, \text{db}) = 100 - (W_1 + W_2 + W_3) \quad (4)$$

Waste Biodegradation Analysis

The biodegradation of the residues was evaluated through analysis of volatile suspended solids (VSS), total organic carbon of the suspended solids (TOC-S), total organic carbon of liquid portion (TOC-L), IC, TKN, and pH. The details of the methodologies used in these analyses have already been presented in the “Waste Characterization” section. All analyses were performed before and after AD. A carbon-based mass balance was conducted to calculate the percentage of carbon conversion throughout the experiment, taking into account the initial total carbon (TC) (converted to mol) and the carbon that left the system in the biogas (CH_4 and CO_2 , in mol) [29].

Biogas Monitoring

Biogas volumetric production was measured every 2 days by water displacement with a device based on the Mariotte principle [30]. To perform the measurement, a hose was connected between the device and the bioreactor. First, the device was depressurized. Then, the bioreactor valve was opened, which resulted in the release of a volume of water equivalent to the volume of biogas generated in the bioreactor. This amount of water was weighed and converted to volume by the density of water at room temperature. The pH of the aqueous solution was kept acid (< 3.0) to avoid the solubilization of some of the biogas components (CO_2 and H_2S). The composition of biogas was accessed weekly through a gas chromatograph (GC-2014 Shimadzu) equipped with a ShinCarbon column (ST 100/120 2 m 1 mm ID 1/16" OD Silco) and TCD detector. Helium was used as the carrier gas at a flow rate of 10 mL/min. The injector and detector temperatures were held at 200 and 250 °C, respectively. The oven program was 40 °C (3 min), ramp at 15 °C/min to 150 °C, and hold for 0.67 min. The results were expressed as percentages of O_2 , N_2 , CH_4 , and CO_2 .

Results and Discussion

Waste Characterization

The initial characteristics of feedstock strongly affect startup, process stability, and biogas production during anaerobic digestion. Table 1 presents the characteristics of the wastes used in this work.

The sludge exhibited high organic carbon content (TOC/TC around 85%), C/N ratio close to 11, pH close to neutral, and a concentration of chromium well below the reported tolerance for the microorganisms involved in the AD process, around 500–5000 ppm [29] and close to the value reported by Zupančič and Jemec [31]. In addition to

Table 1 Characterization of waste used for biogas production

	Chromium (g/L)—sludge; (% (g/g))—shaving and oat straw	TKN (g/L)	TOC (g/L)	IC (g/L)	C/N ratio	VS (%)	pH
Sludge	0.31	1.21	11.27	1.89	10.88	34.42	7.45
Leather shavings	1.14%	2.95%	32.29%	0	10.95	90.24	4.09
Oat straw		0.54%	44.29%	0	82.02	94.10	8.15

TKN total Kjeldahl nitrogen; TOC total organic carbon; IC inorganic carbon; C/N ratio carbon/nitrogen ratio; VS volatile solids

the presence of chromium, the sludge also had a high concentration of vegetable tannins and phenols, probably from the retanning operation, since the tannery from which the sludge was obtained uses chromium as tanning agent. The concentrations of vegetable tannins and phenols in sludge were of 7.83 and 0.96 g/L, respectively. This high concentration is due to an excess of vegetable tannin extracts used during vegetable tanning or retanning process to ensure full penetration and reaction of tannins with hide collagen. Consequently, many tannin components inevitably remain in the wastewater and in the sludge. Thus, added to the toxic phenolic and recalcitrant nature of vegetable tannins, tannin sludge has a high concentration of vegetable tannin and phenols in their composition [32, 33]. Agustini et al. [32] evaluated the AD of sludge obtained from a tannery that uses vegetable tannin as the tanning agent, which has proven to be toxic to AD. The concentration of vegetable tannin present in the sludge used by Agustini et al. (16.52 g/L) was twice as high as the value found in the present work.

Shavings exhibited high organic load, C/N ratio close to observed to sludge (10.95), acidic pH values, and a concentration of chromium of 1.14%. The concentration of chromium in the shavings is due to absorption (fixation) of chromium in the hide during the tanning process [22] and did not impart any toxicity for its AcoD, both in this study and in previous ones [23, 30, 31].

The oat straw exhibited high organic carbon load and thus a high C/N ratio, of 82. This could limit its efficient use in mono-AD processes for methane production. Also, oat straw (raw and pretreated) was characterized in relation to its chemical composition and functional groups and the results will be presented and discussed in the next section.

Despite the complexity of the sludge, the recalcitrant characteristics of all the wastes, and the low C/N ratio of the mixture used (around 13), the AD was established, proven by the production of methane.

Effect of the Pretreatments on Oat Straw

In this study, the effect of different pretreatments applied to oat straw was evaluated with respect to changes in chemical composition and functional groups.

Changes in Main Chemical Composition

The cellulose, hemicellulose, and lignin contents of untreated and pretreated oat straw are shown in Table 2. No correction for ash content was done. Extractives (or soluble matter) include pectins, proteins, fats, and glucuronic and galacturonic acids [11].

The untreated oat straw presented 51.92% of hemicellulose, 24.41% of cellulose, and 19.53% of lignin. This composition is different from that presented by Gomez-Tovar et al. [11] (28.2, 35.0, and 4.1%, respectively), which can be attributed to climatic and geographic conditions, as well as the sample preparation and extraction method [34].

The influence of the pretreatments on lignocellulosic composition of oat straw was also investigated. The extractive content did not show significant difference among the tested cases. After acid, alkaline, and alkaline + thermal pretreatment, increase in cellulose content was accomplished by the decrease in hemicellulose, which is in accordance with the findings of Krishania et al. [35], Rajput et al. [15], and Shen et al. [36]. Cellulose content increased by about 176% due to alkaline + thermal pretreatment as compared

Table 2 Effects of pretreatment on the chemical composition of oat straw. For each column, similar letters a, b, and c represent values that are not significantly different by the Tukey's test ($p > 0.05$)

	Extractives (%)	Hemicellulose (%)	Cellulose (%)	Lignin (%)
Untreated	4.14 ± 0.32 ^a	51.92 ± 0.83 ^a	24.41 ± 1.63 ^b	19.53 ± 0.74 ^b
Acid pretreatment	1.23 ± 0.16 ^a	45.59 ± 0.09 ^c	30.90 ± 1.00 ^b	22.29 ± 1.00 ^b
Alkaline pretreatment	3.83 ± 0.28 ^a	31.26 ± 0.29 ^d	56.37 ± 1.68 ^a	8.54 ± 0.33 ^c
Thermal pretreatment	3.26 ± 0.03 ^a	50.90 ± 0.04 ^{a,b}	24.19 ± 0.46 ^b	21.65 ± 0.30 ^b
Acid + thermal pretreatment	2.22 ± 0.26 ^a	46.46 ± 0.08 ^{b,c}	19.89 ± 1.55 ^b	31.44 ± 0.15 ^a
Alkaline + thermal pretreatment	2.52 ± 0.26 ^a	21.07 ± 0.64 ^c	67.47 ± 1.44 ^a	8.94 ± 0.55 ^c

to untreated oat straw. Ran et al. [37] reported the same increase in cellulose contents due to hydrothermal pretreatment of lignocellulosic biomass at 160 °C. Lignin content decreased by about 55% due to alkaline and alkaline + thermal pretreatments as compared to untreated oat straw. The structural changes caused by these pretreatments can be attributed to the action of the alkali on the ether and ester linkages between hemicellulose and lignin. Due to the breaking of these internal bonds, both hemicellulose dissolution and lignin removal can be achieved, which can contribute to improve the digestibility of the substrate [14, 36].

Changes in Functional Groups

The change in the chemical bonds of oat straw due to pretreatments was analyzed by the FTIR technique. The spectra of pretreated and untreated straw in the 4000–500 cm^{-1} region are shown in Fig. S1.

The spectra of untreated and pretreated oat straw have similar profiles but varied intensities. The infrared spectra showed characteristic bands of the functional groups of the components of the lignocellulosic fibers (cellulose, hemicelluloses, and lignin) [38]. These components showed mainly alkanes, aromatic groups, and different functional groups, such as ester, ketone, and alcohol in their structures [34, 39]. Similar to that reported in the literature [15, 40, 41], all samples presented two main absorbance regions: the first one, in wavelengths in the range of 700–1800 cm^{-1} , and the second one at higher wavelengths corresponding to the range 2800–3500 cm^{-1} .

As can be seen in Fig. S1, the absorption band at 3330 cm^{-1} attributed to the O–H stretching in the hydrogen bond of cellulose, hemicellulose, and lignin [15, 38, 42] decreased to oat straw submitted to thermal (d), alkaline (c), and acid pretreatments (b). This implies that crystalline cellulose of oat straw was disrupted by these pretreatments.

The peak of 2920 cm^{-1} correspondent to C–H_n and related to asymmetric methylene stretching of cellulose, hemicellulose, and lignin becomes weaker with acid pretreatment (b) [41]. For oat straw submitted to thermal (d), acid + thermal (e), alkaline + thermal (f), and alkaline pretreatments (c), a peak formation was observed in 2850 cm^{-1} , also related to the C–H_n stretching and characteristic of hemicellulose compounds [42].

The absorption band at 1640 and 1600 cm^{-1} is originated from lignin, which includes C=C stretching of aromatic ring and aromatic skeleton vibration of C=C, respectively [41]. The intensity of these peaks decreased for the oat straw submitted to acid + thermal (e), alkaline (c), acid (b), and alkaline + thermal (f) pretreatments and is related to the removal of lignin.

Changes in the intensity of the peaks 1370 and 1318 cm^{-1} are detected to acid + thermal (e), alkaline + thermal (f),

alkaline (c), and acid (b) pretreated oat straw. These bands are related to aliphatic C–H deformation vibration and C–H₂ wagging of cellulose, respectively [41].

The stretching vibration of C–O–C for cellulose and hemicellulose is described at the approximate wavenumber of 1158 cm^{-1} [41, 43]. Compared to the untreated oat straw (a), the decrease in wavenumber intensity observed in acid (b) and alkaline pretreatment (b) might be because of the removal of hemicellulose [36].

The absorption band at 1100 cm^{-1} , also related to the vibration of C–O–C glycosidic linkages of the cellulose [34], and 1031 cm^{-1} , related to C–O stretching of cellulose and hemicellulose [41, 42], becomes weaker for alkaline + thermal (f), alkaline (c), and acid pretreatments (b).

The intensity at 899 cm^{-1} band corresponding to β -D-cellulose linkages had a significant decrease in acid pretreatment (b), which indicated the removal of amorphous components [13, 40].

FTIR spectra indicate that pretreatments result in some level of delignification of the fiber. However, some peaks corresponding to lignin components (1740, 1510, 1455, and 1234 cm^{-1}) increased or even were only detected in pretreated oat straw [34, 38, 41]. This contradiction can be attributed to the formation of pseudo-lignin droplets, spherical structures containing a lignin-like compound that can be found on the surface of pretreated biomass [40]. This phenomenon is possible when some pretreatments generate products such as furfural repolymerize and/or lignin and form a lignin-like material named pseudo-lignin [44].

Taken together, these FTIR results indicate that specially the oat straw pretreated with acid, alkaline, and by the combination of alkaline + thermal presented changes in the lignocellulosic structure of the fiber related to solubilization of carbohydrate and lignin, breakage of association between lignin with carbohydrate and changes in aromatic rings. These changes might be beneficial for AD, because the solubilization of lignin can make cellulose fibers become more available for enzyme attack to convert it into fermentable sugars.

Effect of Pretreatment on Biogas Production and Waste Treatment

The cumulative biogas production per gram of VSS added for each experimental condition tested is shown in Fig. 2. The daily biogas production and biogas composition for all the tests are shown in Fig. S2. Waste analyses, expressed by the parameters TOC-L, TOC-S, VSS, IC, TKN, and pH, before and after biodegradation assays, are shown in Tables 3 and 4.

The definition of the acronyms that represent the composition of the reactors is as follows: sludge + shavings (SLU + SHA), sludge + shavings + untreated oat straw

(SLU + SHA + UOS), sludge + shavings + oat straw submitted to acid pretreatment (SLU + SHA + OS-HCl), sludge + shavings + oat straw submitted to alkaline pretreatment (SLU + SHA + OS-NaOH), sludge + shavings + oat straw submitted to thermal pretreatment (SLU + SHA + OS-THE), sludge + shavings + oat straw submitted to acid + thermal pretreatments (SLU + SHA + OS-HCl/THE), and sludge + shavings + oat straw submitted to alkaline + thermal pretreatments (SLU + SHA + OS-NaOH/THE).

No Oat Straw—SLU + SHA

The reactors without the addition of oat straw produced on average 16.17 ± 1.80 mL of cumulative biogas/g of VSS added (12.57 g) with a maximum percentage of 38% of methane (Fig. 2 and Fig. S2). The lag phase of the cumulative biogas production curve was approximately 30 days, showing that the recalcitrance and complexity of the residues caused the hydrolysis phase to take longer to complete. Similarly, the log phase was also long, beginning at day 30 of incubation and ending around day 150, and also not having a large slope, again due to the complexity of the residues, so that the microorganisms could not quickly metabolize the wastes (Fig. 2). A plateau phase was verified in the cumulative biogas production curve, a phenomenon that is also known as diauxie and that could be related to the accumulation of volatile fatty acids (VFAs), indicating an imbalance between the early stages of AD and methanogenesis. The methanogenic archaea are not able to process all the intermediate products so fast and the high concentration of the acids that starts to accumulate reduces the pH of the medium which causes the inhibition of methane production. Biogas production continues when the VFA concentration drops [45, 46]. The methane concentration rose along with the

biogas production (Fig. S2) as expected for an established AD process. Its percentage reached a maximum of 38% at the end of the log phase.

The biogas production and methane yield obtained in this work were lower than that reported in other studies carried out with equivalent wastes, which produced up to 30 mL of biogas/g of VSS, with a methane percentage of 60% [22, 24]. Comparing the wastes used in the studies cited above with that used in the present work, the main difference is in the composition of the sludge. In addition to a higher concentration of chromium, the sludge used in this study contains phenols and vegetable tannin, which have been reported as toxic to the final stages of AD in previous studies [23, 32] and which may have partially inhibited the AD of the residues.

In our study, the efficiency of waste treatment (reducing its organic load) is as important as the production of biogas, since the remaining waste (after the AD process) still needs to be managed and a more mineralized waste is environmentally inert for disposal. It is worth remembering that, in addition to producing biogas, the AD process also generates a nitrogen-rich digestate that can be used as fertilizer in the field, which can be particularly interesting in this case, where one of the wastes comes from agriculture.

In relation to the reduction of the organic load of the wastes, a reduction of 35.13% was observed for VSS (Table 3), which represents the suspended organic matter. Similar to what occurs with VSS, the balance between TOC-S and TOC-L is related to the migration of organic carbon, initially precipitated in the waste and which, due to AD, undergoes hydrolysis and is solubilized in the medium to be metabolized by microorganisms. The variation of TOC-S and TOC-L over the tests is shown in Table 3. For the SLU + SHA test, a reduction in TOC-S (−39.33%) was observed, which is in accordance with the discussion above,

Fig. 2 Biogas cumulative production per gram of VSS added in the biodegradation reactors

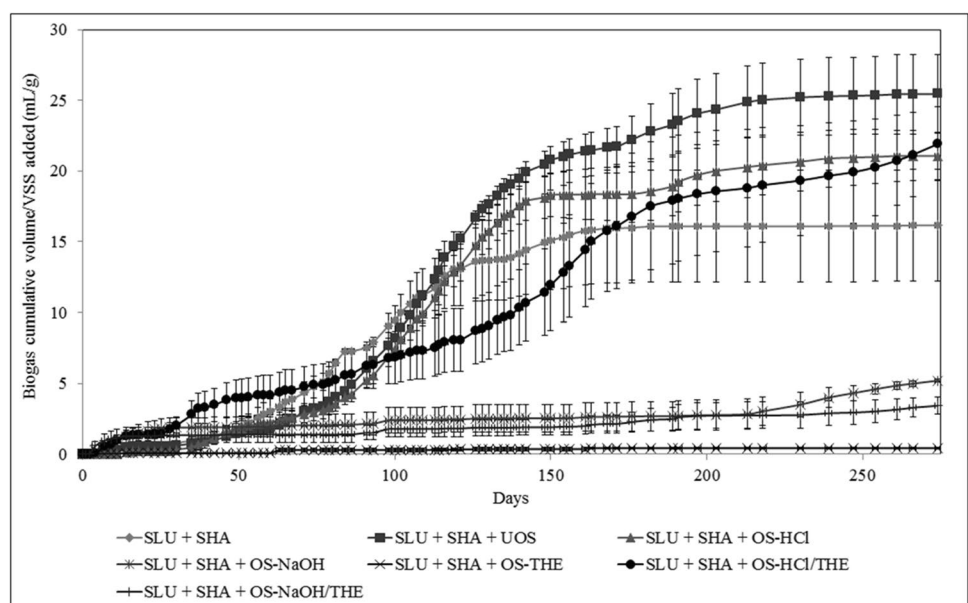


Table 3 Mean values of organic matter (TOC and VSS) of bioreactors before and after biodegradation and variation due to biodegradation. For each column, similar letters a and b represent values that are not significantly different by the Tukey’s test ($p > 0.05$)

Reactor	TOC-L (mg/L)			TOC-S (%)			VSS (%)		
	Before	After	Variation	Before	After	Variation	Before	After	Variation
SLU + SHA	289.80 ± 4.40	399.82 ± 51.09	+ 39.16 ± 19.74% ^b	23.88 ± 0.74	14.43 ± 0.01	− 39.33 ± 1.90% ^b	48.33 ± 0.86	31.29 ± 0.29	− 35.13 ± 1.75% ^a
SLU + SHA + UOS	428.80 ± 40.30	562.60 ± 29.10	+ 38.65 ± 19.82% ^b	28.86 ± 0.52	22.6 ± 0.20	− 21.62 ± 0.73% ^b	70.36 ± 2.30	60.92 ± 2.59	− 13.52 ± 0.85% ^a
SLU + SHA + OS-HCl	323.60 ± 4.20	449.90 ± 9.25	+ 38.97 ± 1.05% ^b	27.53 ± 0.27	33.77 ± 0.32	+ 22.76 ± 2.36% ^a	67.34 ± 1.21	61.25 ± 0.89	− 8.83 ± 2.96% ^a
SLU + SHA + OS-NaOH	314.10 ± 13.35	2509.82 ± 51.09	+ 702.08 ± 17.82% ^a	29.97 ± 1.86	28.21 ± 1.19	− 5.39 ± 1.89% ^{a,b}	72.00 ± 2.06	49.32 ± 3.15	− 31.78 ± 2.43% ^a
SLU + SHA + OS-THE	285.80 ± 5.50	501.00 ± 45.20	+ 76.78 ± 19.22% ^b	29.74 ± 1.35	21.33 ± 0.52	− 28.02 ± 1.52% ^b	65.90 ± 0.04	62.28 ± 2.92	− 5.49 ± 4.50% ^a
SLU + SHA + OS-HCl/ THE	302.50 ± 6.25	732.70 ± 119.15	+ 145.89 ± 44.47% ^b	32.20 ± 1.28	27.46 ± 0.67	− 13.83 ± 5.50% ^b	67.35 ± 1.04	54.44 ± 1.49	− 18.95 ± 4.50% ^a
SLU + SHA + OS-NaOH/ THE	356.40 ± 20.50	1970.50 ± 52.75	+ 463.79 ± 47.23% ^a	29.88 ± 0.10	23.55 ± 0.72	− 21.13 ± 2.66% ^b	68.31 ± 0.38	46.30 ± 0.99	− 32.19 ± 1.83% ^a

Table 4 Mean values of IC, TKN, and pH of bioreactors before and after biodegradation and variation due to biodegradation. For each column, similar letters a and b represent values that are not significantly different by the Tukey’s test ($p > 0.05$)

Reactor	IC (mg/L)			TKN (mg/L)			pH		
	Before	After	Variation	Before	After	Variation	Before	After	Variation
SLU + SHA	198.36 ± 9.32	872.90 ± 8.55	+ 344.80 ± 25.21% ^a	207.2 ± 14.00	644.00 ± 19.60	+ 213.99 ± 11.76% ^a	8.30 ± 0.06	7.79 ± 0.04	− 6.08 ± 0.11% ^a
SLU + SHA + UOS	336.30 ± 20.75	933.30 ± 13.25	+ 182.80 ± 21.39% ^a	201.60 ± 0.00	663.60 ± 15.40	+ 229.17 ± 7.64% ^a	7.86 ± 0.33	7.87 ± 0.06	+ 0.68 ± 3.34% ^a
SLU + SHA + OS-HCl	274.00 ± 21.10	857.03 ± 25.68	+ 217.43 ± 15.07% ^a	207.20 ± 2.80	556.73 ± 22.63	+ 168.30 ± 7.30% ^a	7.67 ± 0.15	7.80 ± 0.09	+ 1.94 ± 3.10% ^a
SLU + SHA + OS-NaOH	222.87 ± 13.66	695.90 ± 84.85	+ 226.49 ± 58.09% ^a	176.40 ± 1.40	613.20 ± 14.00	+ 247.45 ± 5.18% ^a	7.68 ± 0.13	6.93 ± 0.06	− 9.61 ± 2.31% ^a
SLU + SHA + OS-THE	305.70 ± 3.25	862.20 ± 23.20	+ 182.49 ± 10.59% ^a	198.80 ± 1.40	473.20 ± 64.40	+ 137.16 ± 30.72% ^a	7.53 ± 0.02	7.75 ± 0.02	+ 2.92 ± 0.01% ^a
SLU + SHA + OS-HCl/ THE	274.70 ± 1.55	957.30 ± 16.45	+ 248.40 ± 4.02% ^a	182.00 ± 9.80	618.80 ± 5.60	+ 243.32 ± 15.41% ^a	7.65 ± 0.01	8.06 ± 0.06	+ 5.36 ± 0.71% ^a
SLU + SHA + OS-NaOH/THE	255.97 ± 43.61	654.80 ± 7.80	+ 187.07 ± 45.87% ^a	190.40 ± 5.60	641.20 ± 28.00	+ 236.20 ± 4.82% ^a	8.00 ± 0.26	7.02 ± 0.02	− 11.86 ± 2.59% ^a

Table 5 Carbon-based mass balance. For each column, similar letters a and b represent values that are not significantly different by the Tukey's test ($p < 0.05$)

Reactor	Initial C amount ($\times 10^{-2}$ g)	Initial C ($\times 10^{-2}$ mol)	C ($\times 10^{-2}$ mol) CH ₄	C ($\times 10^{-2}$ mol) CO ₂	C conversion
SLU + SHA	34.86 $\pm$ 0.43	2.90 $\pm$ 0.04	0.17 $\pm$ 0.05	0.34 $\pm$ 0.05	17.25 $\pm$ 3.37% ^a
SLU + SHA + UOS	74.93 $\pm$ 0.33	6.24 $\pm$ 0.03	0.51 $\pm$ 0.06	1.17 $\pm$ 0.09	26.80 $\pm$ 2.46% ^a
SLU + SHA + OS-HCl	68.50 $\pm$ 0.04	5.71 $\pm$ 0.003	0.29 $\pm$ 0.03	0.94 $\pm$ 0.04	21.54 $\pm$ 1.31% ^a
SLU + SHA + OS-NaOH	72.01 $\pm$ 03.11	6.00 $\pm$ 0.26	0.02 $\pm$ 0.008	0.22 $\pm$ 0.004	4.17 $\pm$ 0.11% ^b
SLU + SHA + OS-THE	72.79 $\pm$ 2.49	6.07 $\pm$ 0.21	0.003 $\pm$ 0.0002	0.02 $\pm$ 0.004	0.36 $\pm$ 0.07% ^b
SLU + SHA + OS-HCl/THE	77.37 $\pm$ 2.45	6.45 $\pm$ 0.20	0.30 $\pm$ 0.05	0.98 $\pm$ 0.10	20.24 $\pm$ 3.13% ^a
SLU + SHA + OS-NaOH/THE	73.53 $\pm$ 1.24	6.13 $\pm$ 0.10	0.009 $\pm$ 0.002	0.11 $\pm$ 0.009	2.00 $\pm$ 0.21% ^b

and an increase in TOC-L (+39.16%), which not only shows that the suspended carbon was hydrolyzed and dissolved in the medium, but also indicates that the following steps (acidogenesis, acetogenesis, and methanogenesis) were partially inhibited.

The concentrations of TKN and IC increased throughout the process (Table 4), as expected [24, 29]. The increase in TKN concentration (+213.99%) is expected due to the natural characteristic of the waste, which, because of the animal hide origin, is a nitrogen-rich source. As there was no large outflow of gaseous nitrogen, it remained in the residues and increased its concentration due to the volumetric reduction of the system. The IC concentration also increased (+344.80%), as expected due to the mineralization of the waste.

Untreated Oat Straw—SLU + SHA + UOS

The AcoD between tannery waste (shavings + sludge) and untreated oat straw (SLU + SHA + UOS) produced on average 25.44 $\pm$ 0.69 mL of cumulative biogas/g of VSS added (19 g), which represents an increase of almost 60% in relation to the case discussed above, without the addition of oat straw. The periods comprising the lag and log phases were practically the same as those observed for the SLU + SHA. However, a greater slope of the curve was observed for SLU + SHA + UOS, indicating that the wastes were more easily metabolized in this case, probably due to the greater nutritional balance provided by the addition of oat straw (C/N ratio of 13, closest to optimal for AD) (Fig. 2). As in the case without the addition of oat straw, discussed above, the cumulative production of biogas also showed diauxic growth curves. The concentration of CH₄ increased with the increase in biogas production, reaching the maximum percentage of 40% at the end of the log phase (Fig. S2).

Regarding the treatment of waste (Tables 4 and 5), the parameters analyzed follow the same trend as observed for SLU + SHA. Reduction of VSS (−13.52%) and TOC-S (−21.62%) and increase in TOC-L (+38.65%) indicate a

decrease in organic matter and the migration of organic carbon from the suspended to the dissolved phase, with partial inhibition of the steps subsequent to hydrolysis. Also, the concentrations of TKN and IC increased (+229.17% and +182.80%, respectively) throughout the process, as expected.

Pretreated Oat Straw—SLU + SHA + OS-HCl; SLU + SHA + OS-NaOH; SLU + SHA + OS-THE; SLU + SHA + OS-HCl/THE; SLU + SHA + OS-NaOH/THE

The cumulative production of biogas per gram of added VSS for the cases in which pretreated oat straw was added to the bioreactors was as follows: 21.06 $\pm$ 0.99 mL/g VSS (18.18 g) for SLU + SHA + OS-HCl; 5.20 $\pm$ 0.74 mL/g VSS (19.44 g) for SLU + SHA + OS-NaOH; 21.91 $\pm$ 2.06 mL/g VSS (18.19 g) for SLU + SHA + OS-HCl/THE; and 3.43 $\pm$ 0.58 mL/g VSS (18.44 g) for SLU + SHA + OS-NaOH/THE. The test in which the oat straw was submitted only to the thermal pretreatment (SLU + SHA + OS-THE) produced virtually no biogas (0.42 $\pm$ 0.09 mL/g VSS), being statistically not different ($p > 0.05$) from SLU + SHA + OS-NaOH and SLU + SHA + OS-NaOH/THE.

The curve profile (lag and log phases and slope of the curve) of the SLU + SHA + OS-HCl case is very similar to that observed for SLU + SHA + UOS. For the SLU + SHA + OS-HCl/THE case, although biogas production started earlier (lag phase started around the 10th day), the slope of the curve is much lower (Fig. 2), indicating that the combination of acid and thermal pretreatments was an aggressive process, which sterilized the wastes and caused an excessive degradation of the organic matter that would later be digested [32]. For both cases, diauxic behavior was observed, which, as mentioned previously, could be related to the VFA concentration [45, 46].

The percentage of methane showed the same growth as the biogas production growth for SLU + SHA + OS-HCl and SLU + SHA + OS-HCl/THE, reaching a maximum of 40.85% and 30.31%, respectively, both at the end

of the process (Fig. S2), showing that anaerobic digestion took longer to establish and did not establish so well. For SLU + SHA + OS-NaOH and SLU + SHA + OS-NaOH/THE, the percentage of methane reached was 17.46% and 10%, respectively. For thermal pretreatment (SLU + SHA + OS-THE), the percentage of methane reached 18% in the process (Fig. S2), showing that even though AD was not fully established, methanogenic activity did occur. The high concentration of CO₂ (up to 70%) observed for all cases corroborates the hypothesis that mainly the final stage of AD was inhibited. It means, the methanogenic bacteria were not able to metabolize the CO₂ formed during the initial stages and convert it into methane; however, acetogenesis was established [47, 48].

The amount of biogas produced for these cases is lower than that observed for the case in which the oat straw was not pretreated (SLU + SHA + UOS) (25.44 mL of cumulative biogas/g of VSS added), suggesting that pretreatment probably generated inhibitory compounds. This inhibition can be attributed to the presence of toxic compounds from dehydration of pentoses (furfural) and hexoses (hydroxymethyl furfural) and/or to phenolic compounds from the rupture of lignin, which inhibit anaerobic fermentation because it is difficult for anaerobic microorganisms to metabolize them [6, 11, 37]. Particularly for cases involving thermal pretreatment, the observed inhibition can also be due to autoclaving being a very aggressive process, inducing much sterilization and excessive degradation of the organic matter that would be further digested [32].

Lower biogas production and methane yields observed for cases where the oat straw was pretreated with NaOH can also be attributed to the presence of Na⁺ ions. Sodium is a known process inhibitor in anaerobic systems and impacts on methanogens through an increase of osmotic pressure or complete dehydration of microorganisms [5, 49]. Although pretreatment with NaOH has a positive effect on a series of studies reported involving AD of lignocellulosic materials [7, 14, 16], and even in the lignin removal observed for this study (Table 2), it is believed that the possible presence of ammonia due to the nature of the leather wastes, along with Na⁺ ions originating from the pretreatment of the straw, may have caused the observed inhibition. This hypothesis is supported since Hierholtzer and Akunna [50] found that relatively low sodium levels (0.2 mol/L) can bring about significant levels of process inhibition in the presence of high levels of ammoniacal nitrogen. In addition, the pH values observed after biodegradation, 6.93 for SLU + SHA + OS-NaOH and 7.02 for SLU + SHA + OS-NaOH/THE (Table 4), well below the average observed for the other cases, are an indicative that the acidogenesis was established, but the products were not converted into CH₄ and CO₂ by inhibition of methanogenesis, corroborating the assumption made above.

Regarding the efficiency in the treatment of the wastes, a reduction in VSS (Table 3) was observed in all cases containing pretreated oat straw as a co-substrate. The migration of organic carbon (TOC-S to TOC-L), which was initially precipitated in the residue and, after hydrolysis, is solubilized in the medium, was also observed for the cases discussed in this section. The values of TOC-L for SLU + SHA + OS-NaOH and SLU + SHA + OS-NaOH/THE increased more than 400% (Table 3). These results indicate the toxicity of the wastes to the final stages of AD, since only the hydrolysis, which releases organic matter in the medium, was able to establish [23]. The concentrations of TKN and IC increased after AD for all cases, proving the mineralization of the waste, even though AD has not been fully established in some cases.

Reduction in TOC and VSS values and increases in IC and TKN values are common for cases where the AD process was well established, with satisfactory production of biogas and methane. For SLU + SHA + OS-THE, even with practically no biogas production, there was a reduction in VSS (− 5.49%) and TOC-S (− 28.02%) and an increase in TKN (+ 137.16%) and IC (+ 182.49%). This behavior can be attributed to the activity of microorganisms not involved in AD, so that the mineralization of the wastes is not only linked to the biogas production.

Carbon-Based Mass Balance

A balance that involves TC (TOC + IC) was carried out for the mixtures in bioreactors in order to compare the amount of initial carbon with the amount of carbon that was converted to CH₄ and CO₂ throughout the experiment. The results are detailed in Table 5, where it is shown that the AcoD between tannery waste (shavings + sludge) and untreated oat straw was the most efficient, converting almost 27% of carbon from the waste to CH₄ and CO₂. The AcoD of tannery wastes (shavings + sludge) converted 17%, the same carbon conversion observed by Agustini et al. [29] in their study involving the AcoD of leather wastes similar to those used in this work. Among the pretreatments tested on oat straw, acid pretreatment stood out as the most efficient option, converting 21% of carbon from the waste into CH₄ and CO₂.

The greater efficiency in carbon conversion observed for SLU + SHA + UOS highlights the importance of the AcoD process. In fact, the balance of nutrients provided by AcoD has resulted in a more efficient process with greater biogas production, in addition to the economic advantage of sharing the AD system and treating three different wastes at the same time. Although the pretreatments studied here have improved the characteristics and bioavailability of oat straw, there is still the possible formation of secondary products or sterilization of important microorganisms that did not reflect in a greater production of biogas.

Conclusion

The experiments of AcoD of sludge, shavings, and oat straw (untreated and pretreated) indicate that the following: (1) the addition of untreated oat straw improved the AcoD process, increasing biogas production by almost 60% when compared to the AcoD of only leather waste; (2) acid and acid + thermal pretreatments of oat straw did not enhance biogas and methane production compared to untreated oat straw; (3) lignin content decreased by about 55% due to alkaline and alkaline + thermal pretreatments as compared to untreated oat straw, but the presence of inhibitory compounds remaining of NaOH pretreatment along with the presence of ammoniacal nitrogen due to the nature of the leather wastes may have partially inhibited methanogenesis, limiting the production of biogas for these cases; (4) thermal pretreatment of oat straw probably have generated toxic compounds from hemicellulose and cellulose degradation, which inhibited the AcoD process and, as consequence, virtually no biogas was produced.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12155-021-10285-1>.

Data Availability Not applicable.

Code Availability Not applicable.

Declarations

Competing Interests The authors declare no competing interests.

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